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EDUCATION

B.S., Peking University, Beijing, China

Sep 2015 – July 2019

School of Earth and Space Sciences

Thesis Topic: Radiative Hydrodynamics Modeling of Eruptive Events on the Sun and M Dwarfs

Ph.D., University of Michigan, Ann Arbor, MI

Sep 2019 – Dec 2023

Department of Climate and Space Sciences and Engineering

Dissertation Topic: New Insights into the Coronal Heating Problem: Analysis of Spectral Line Widths in the Solar Corona

EXPERIENCE

Peking University, Beijing, China

Apr 2017 – July 2019

Undergraduate research supervised by Dr. Hui Tian

- Investigation of an S-shape Flare Ribbon with SDO and NVST observations
- Spectroscopic Diagnostics of IRIS data

National Solar Observatory, Boulder, CO

July 2018 – Sep 2018

Summer research supervised by Dr. Adam F. Kowalski

- Radiative Hydrodynamics Modeling of Mg II Lines at Solar Flare Ribbons

University of Michigan, Ann Arbor, MI

Sep 2019 – Dec 2023

Advisor: Dr. Enrico Landi

- Spectral Line Widths at Polar Off-limb Corona Observed by Hinode/EIS
- Ion Temperatures in a Polar Coronal Hole Observed by Hinode/EIS and SOHO/SUMER

Institute for Astronomy, University of Hawaii, Honolulu, HI

July 2022

Short visiting working with Dr. Shadia Habbal and Dr. Adalbert Ding

- Spectroscopic Observations during the 2017 Total Solar Eclipse

Physikalisch-Meteorologische Observatorium Davos/World Radiation Center (PMOD/WRC), Switzerland

Jan 2024 – Present

Postdoctoral Researcher supervised by Prof. Louise Harra

- Upflows at Active Region Boundaries

Institute for Particle Physics and Astrophysics, ETH Zürich

Jan 2025 – Present

Postdoctoral Researcher supervised by Prof. Louise Harra

- Upflows at Active Region Boundaries

PUBLICATIONS

Zhu, Y., Harra, L. K., Barczynski, K., et al. “Active region upflows in various coronal structures and their coupling to the lower atmosphere”, 2025, [A&A](#), **701**, A205

Zhu, Y., Habbal, S. R., Adalbert, D., Yamashiro, B., Landi, E., Benjamin, B., Constantinou, S., Nassir, M., “Spectroscopic Observations of the Solar Corona during the 2017 August 21 Total Solar Eclipse: Comparison of Spectral Line Widths and Doppler Shifts between Open and Closed Magnetic Structures”, 2024, [ApJ](#), **966**, 122

Zhu, Y., Szente, J., Landi, E., “Estimating Ion Temperatures at the Polar Coronal Hole Boundary”, 2023, [ApJ](#), **948**, 90

Zhu, Y., Szente, J., Landi, E., “Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii”, 2021, [ApJ](#), **913**, 74

Zhu, Y., Kowalski, A. F., Tian, H., Uitenbroek, H. *et al.* “Modeling Mg II h, k and Triplet Lines at Solar Flare Ribbons”, 2019, [ApJ](#), **879**, 19.

Barczynski, K., Janvier, M., Nelson, C. J., Schad, T., Tritschler, A., Harra L. K., Müller, D., Parenti, S., Valori, G., Cauzzi, G., **Zhu, Y.**, “First coordinated observations between Solar Orbiter and the Daniel K. Inouye Solar Telescope”, [A&A](#), 2025, **701**, A77.

Huang, Z., Velli, M., Shi, Chen., **Zhu, Y.**, et al. “Dominance of 2 Minute Oscillations near the Alfvén Surface”, [ApJL](#), 2024, **977**, 12.

Yang, Z., Tian, H., **Zhu, Y.**, Xu, Y., Chen, L., Sun, Z., “Is It Possible to Detect Coronal Mass Ejections on Solar-type Stars through Extreme-ultraviolet Spectral Observations?”, [ApJ](#), 2024, **966**, 24.

Yang, Z., Tian, H., Bai, X., Chen, Y., Guo, Y., **Zhu, Y.** *et al.* “Can We Detect Coronal Mass Ejections through Asymmetries of Sun-as-a-star Extreme-ultraviolet Spectral Line Profiles?”, [ApJS](#), 2022, **260**, 36.

ACADEMIC AWARD

Outstanding Student Presentation Award (OSPA), AGU 2022 Fall Meeting	2022
Outstanding Student Poster Award, 2 nd China-Europe Solar Physics Meeting	2019

ACADEMIC ACTIVITIES

Referee for

- Astronomy & Astrophysics (A&A) and The Astrophysical Journal (ApJ)

Teaching

- **Fall 2025:** TA leading exercise classes of the Physics of Space Weather.

Talks

- “Chromospheric Fibrils in Mossy Active Region Plage Observed by DKIST, IRIS, and Solar Orbiter”, Hinode-18/IRIS-16 June 2025
- “Upflows in a Decaying Active Region and its Potential Contribution to the Solar Wind”, EGU 2025 April 2025
- “Identifying Potential AR Upflow Regions in SPICE Doppler maps as Counterparts to Upflows Observed by Hinode/EIS: Preliminary Results with PSF Deconvolution Efforts”, SPICE Consortium meeting June 2024
- “Spectroscopic Observations of the Solar Corona in Different Wavelengths and from Multiple Vantage Points”, Joint Solar Orbiter, Parker Solar Probe, and DKIST Meeting 2024 April 2024

- “Spectroscopic Observations of the Corona in the Visible and Extreme Ultraviolet during the 2017 August 21 Total Solar Eclipse”, SPD 2023 Aug 2023
- “Spectroscopic Observations of the Corona in the Visible and Extreme Ultraviolet during the 2017 August 21 Total Solar Eclipse”, AGU 2022 Fall Meeting Dec 2022
- “Ion Temperature Diagnostics at the Off-limb Coronal Hole Boundary and Post-flare Loops”, TESS 2022 Meeting Aug 2022
- “Modeling Mg II h , k and Triplet Lines at Solar Flare Ribbons”, ISSI/ISSI-BJ International Team Workshop: Diagnosing Heating Mechanism in Solar Flares Through Spectroscopic Observations of Flare Ribbons Oct 2018
- “An S-shape Flare Ribbon Observed by NVST and SDO/AIA”, The 2017 Annual Meeting of Chinese Geoscience Union (CGU) Oct 2017

Posters

- “Upflows with different morphologies at active region boundaries observed by Hinode and Solar Orbiter”, ESPM-17 Sep 2024
- “Heating of Heavy Ions in Active Regions Observed by SUMER: Flares, Loop brightening, and Coronal Rain”, AGU 2023 Fall Meeting Dec 2023
- “Spectroscopic Observations of the Solar Corona in the Visible, Extreme Ultraviolet, and Near-Infrared during the 2017 Total Solar Eclipse”, Hinode-16/IRIS-13, Niigata, Japan Sep 2023
- “Active Region Heating: Flares, Loop brightening, and Coronal Rain”, SHINE Workshop 2023, Aug 2023
- “Estimating Ion Temperatures at the Polar Coronal Hole Boundary”, SHINE Workshop 2022, June 2022
- “Ion Temperature Diagnostics at the Off-limb Coronal Hole Boundary and Post-flare Loops”, AGU 2021 Fall Meeting Dec 2021
- “Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii”, Hinode-14/IRIS-11 Meeting Oct 2021
- “Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii”, SHINE 2021 Jul 2021
- “Fe XII and Fe XIII Line Widths in the Polar Off-limb Solar Corona up to 1.5 Solar Radii”, AGU 2020 Fall Meeting Dec 2020
- “Modeling Mg II h , k , and Triplet Lines at Solar Flare Ribbons”, The 2nd China-Europe Solar Physics Meeting May 2019
- “An S-shape Flare Ribbon Observed by NVST and SDO/AIA”, The 4th Chinese Space Weather Conference Aug 2017